

Data Engineering Portfolio

Senior Data Engineer specializing in Azure Databricks, PySpark, Delta Lake, Kafka Streaming, and Cloud-Scale Data Platforms. 7+ years building production-grade pipelines for enterprise-scale workloads.

- Azure Databricks
- PySpark
- Delta Lake
- Apache Kafka
- AutoLoader
- Unity Catalog
- ADLS Gen2
- Spark Structured Streaming
- Uber H3
- Machine Learning

ProjectsSkillsTimeline

Featured Projects

Production-grade data engineering POCs showcasing real-world Databricks patterns

2025 – PRESENT

Geo-Location Intelligence & Advertisement Campaign Platform

Built an end-to-end geospatial analytics platform on Azure Databricks using Uber H3 hexagonal indexing, Voronoi cell-site mapping, POI enrichment, and AutoLoader mediation ingestion to enable location-based advertisement campaign analytics.

AutoLoaderUber H3Voronoi

SCD Type 2Delta LakeUnity Catalog

PhotonAQELiquid Clustering

ArchitectureCode Files

2024 – 2025

Real-Time Kafka Streaming Pipeline (Multiplexer Pattern)

Designed and operated a real-time streaming pipeline using Apache Kafka and Spark Structured Streaming. Raw events land as Parquet on ADLS Gen2, stream into a centralised Multiplexer Delta table (topic | target_table | INSERT/UPDATE/DELETE), then a cost-optimised scheduled batch routes records to SCD Type 1 or SCD Type 2 downstream tables based on a metadata routing config.

Apache KafkaSpark Structured Streaming

Delta LakeSCD Type 1 & 2Schema Evolution

Data QualityADLS Gen2

ArchitectureCode Files

2022 – 2023

AutoLoader File-Notification Multiplexer Pipeline

Engineered a metadata-driven AutoLoader pipeline where files landing on ADLS Gen2 trigger Azure Event Grid notifications queued to Azure Queue Storage. AutoLoader (cloudFiles, file-notification mode) streams records into a centralised Multiplexer Delta table. Dedicated streaming jobs per dataset then apply SCD Type 1 or SCD Type 2 to downstream Silver tables in near-real-time.

AutoLoader File-NotificationMetadata-Driven

Delta Tracking TablesSchema Evolution

SCD1 / SCD2Databricks Workflows

ArchitectureCode Files

2021 – 2022

Enterprise Multi-Source ETL Pipeline Platform

Delivered end-to-end extraction and transformation of transactional data from multiple enterprise source systems (SFTP, SharePoint, APIs) into ADLS Gen2 curated Delta tables, with PGP encryption, ADF orchestration, and operational dashboards.

Azure Data FactoryDatabricks Workflows

SFTP + PGPDelta TablesADLS Gen2

Power BI

ArchitectureCode Files

2019 – 2021

OPEX Analytics, Geospatial Dashboards & Contract Cost Analysis

Ingested and analyzed infrastructure OPEX datasets on Azure Databricks, applied business rules for cost attribution, built geospatial Power BI dashboards using Uber H3 indexing, and built a metadata-driven PDF extraction framework to create a structured invoicing dataset — then joined it with the power supply asset register to compare theoretical contract costs against actual invoiced amounts and identify contracts for recalculation.

Azure DatabricksUber H3

PDF Regex ExtractionContract Cost Analysis

PySparkPower BICost Attribution

ArchitectureCode Files

Technical Skills

Core competencies across the modern data engineering stack

CLOUD & STORAGE

▸ Azure Databricks

▸ ADLS Gen2

▸ Azure Data Factory

▸ Azure Event Hubs

▸ Azure Key Vault

▸ Unity Catalog

STREAMING & INGESTION

▸ Apache Kafka

▸ Spark Structured Streaming

▸ Databricks AutoLoader

▸ File Notification Mode

▸ Checkpointing

▸ Schema Evolution

DATA LAKE & DELTA

▸ Delta Lake

▸ Medallion Architecture

▸ SCD Type 1 & 2

▸ Liquid Clustering

▸ Z-Ordering

▸ Delta Time Travel

LANGUAGES & FRAMEWORKS

▸ PySpark

▸ Python

▸ SQL / SparkSQL

▸ FastAPI

▸ React JS

▸ Pandas / NumPy

GEOSPATIAL

▸ Uber H3 Indexing

▸ Voronoi Diagrams

▸ POI Enrichment

▸ OpenStreetMap

▸ Geospatial Joins

▸ Spatial Analytics

ML & ANALYTICS

▸ Scikit-learn / MLlib

▸ Contract Cost Analysis

▸ PDF Regex Extraction

▸ Feature Engineering

▸ Power BI

▸ Data Quality Rules

PERFORMANCE & OPS

▸ Adaptive Query Execution

▸ Photon Engine

▸ Spark Tuning

▸ Databricks Workflows

▸ Job Orchestration

▸ SMTP Alerting

GOVERNANCE & SECURITY

▸ Unity Catalog RBAC

▸ ADLS Gen2 Encryption

▸ PGP Encryption

▸ Data Lineage

▸ PII Compliance

▸ Column-Level Security

Project Timeline

Career progression and key deliverables

2025 – Now

Senior Data Engineer — Geo Intelligence Platform

AutoLoader mediation ingestion, Uber H3 geospatial processing, SCD2, Photon/AQE optimization, Unity Catalog governance

2024 – 2025

Senior Data Engineer — Real-Time Kafka Streaming

Kafka + Spark Structured Streaming, Bronze/Silver/Gold medallion, metadata-driven reusable framework

2022 – 2023

Data Engineer — AutoLoader Metadata Framework

File-notification AutoLoader, Delta tracking tables, schema evolution, 70%+ compute cost reduction

2021 – 2022

Data Engineer — Enterprise ETL Platform

ADF + Databricks, SFTP/PGP ingestion, transactional data curated Delta tables, real-time operational reporting

2019 – 2021

Data Engineer — OPEX Analytics & Contract Cost Analysis

OPEX cost attribution, Uber H3 dashboards, metadata-driven PDF extraction, invoicing dataset, contract cost recalculation